

Lecture 1

24 March 2026 17:35

Simple linear regression

Note that winsorising means replacing outliers with the next lowest/highest value which is not an anomaly. Number of data points stays the same.

Trimming is simply deleting the anomalies/not using them.

When analysing data you can winsorize it, change to log scale, then write each variable in logs, then demean each variable (so that the mean is 0).

If you have demeaned the data, the line of best fit goes through the origin.

For simple linear regression, we have:

$$Y_i = \beta \cdot X_i + \varepsilon$$

Can also write it as:

$$E(Y_i | X_i) = \beta \cdot X_i$$

1. y_i is the **dependent variable**.
2. x_i is the **explanatory variable** or **regressor** (please, not 'independent variable'!).
3. β is the **parameter, coefficient** or **slope**.
4. ε is the **residual** or **error** (please, not 'noise'!).
5. We index observations by $i \in \{1, \dots, N\}$.
6. $\hat{\beta}$ is the parameter estimate, so...
7. $\hat{y}_i$ is the predicted outcome for individual i .

SSR: $\hat{\varepsilon}_i \equiv y_i - \hat{y}_i$

equivalent SSR: $S(\hat{\beta}) = \sum_{i=1}^N \hat{\varepsilon}_i^2$

and that $\hat{\beta} = \frac{\hat{\text{Cov}}(X, Y)}{\hat{\text{var}}(X)}$

where $\hat{\beta}$ minimises SSR.

you'll also find: $\hat{\beta} = \frac{\sum_{i=1}^N x_i \cdot y_i}{\sum_{i=1}^N x_i^2}$

$$\sum_i x_i^2$$

The sample covariance between

The sample covariance between $\hat{\varepsilon}_i$ and x_i is:

$$\widehat{\text{Cov}}(\hat{\varepsilon}, x) \equiv \frac{1}{N} \cdot \sum_{i=1}^N \hat{\varepsilon}_i \cdot x_i = \frac{1}{N} \cdot \sum_{i=1}^N x_i \cdot y_i - \hat{\beta} \cdot \frac{1}{N} \cdot \sum_{i=1}^N x_i^2 = 0$$

$$\therefore \hat{\beta} = \frac{\sum_{i=1}^N x_i \cdot y_i}{\sum_{i=1}^N x_i^2}$$

$\text{Cov}(\varepsilon, x) = (1/N) \cdot \sum (\varepsilon_i - \bar{\varepsilon})(x_i - \bar{x})$ standardly but in the above, in the first expression, it has been demeaned so the mean has already been subtracted. Then just split the sum.

Note that covariance also: $\text{Cov}(x, y) = E[XY] - E[X]E[Y]$

1/N for the population, 1/(N-1) when you only have a sample, which is all real world contexts. Dividing by N-1 makes the sample unbiased.

The image above means that the covariance between x and ε should be zero after fitting, because if there was any covariance left, then x still contains information that we haven't used.

3. Large outliers are unlikely: $0 < \mathbb{E}(\varepsilon_i^4) < \infty$ and $0 < \mathbb{E}(x_i^4) < \infty$.

Just means that the fourth moment (fat-tails) are unlikely.

Simple linear regression sample distribution:

1. Simple Linear Regression is unbiased: $\mathbb{E}(\hat{\beta}) = \beta$;
2. Simple Linear Regression is consistent $\text{plim } \hat{\beta} = \beta$;
3. Simple Linear Regression converges to a Normal distribution: $\hat{\beta} \rightarrow^d \mathcal{N}(\beta, \sigma_\beta^2)$.

Unbiased means on average, across a large enough sample, our estimate hits the true value. I.e, if we repeated our study with many different samples, the average of our β would equal the true β .

Consistent means as our sample size tends to infinity, $\hat{\beta}$ converges to the true β .

The difference between unbiasedness and consistency is that unbiased is about the average across many samples and consistent is what happens as one sample gets bigger.

An analogy:

Imagine you're trying to hit a dartboard (the true β).

- **Unbiased** means: if you threw 1,000 darts, they'd cluster *around* the bullseye on average — even if each individual throw is off
- **Consistent** means: the more you *practice* (bigger n), the closer each individual throw gets to the bullseye

So OLS is consistent, unbiased and Gauss-Markov (BLUE) -- no other linear unbiased estimator has lower variance.

Has asymptotic normality: If we assume the noise has constant variance σ^2 , then as n grows:

$\sqrt{n}(\hat{\beta} - \beta) \rightarrow N(0, \sigma^2 E[XX']^{-1})$ this makes more sense in the later slides and in ML.

i.e. the estimator becomes normally distributed around the truth as you get more data.

CLT means that even if the underlying data isn't normally distributed, β itself will be approximately normally distributed in large samples. It's what makes hyp testing and confidence intervals possible.

A limitation to this is when we only have small samples and the data is heavily skewed or fat-tailed, t-stats and p-values become unreliable.

From here, you can try bootstrapping which is resampling the data many times to empirically build the distribution of β rather than assuming it.

$$\begin{aligned}\hat{\beta} &= \beta + \frac{\sum_{i=1}^N x_i \cdot \varepsilon_i}{\sum_{i=1}^N x_i^2} \\ \therefore \sigma_{\hat{\beta}}^2 &\equiv \text{Var}(\hat{\beta}) = \text{Var}(\beta) + \frac{\sum_{i=1}^N x_i^2 \cdot \text{Var}(\varepsilon_i)}{\left(\sum_{i=1}^N x_i^2\right)^2} = \sigma^2 \cdot \frac{\sum_{i=1}^N x_i^2}{\left(\sum_{i=1}^N x_i^2\right)^2} \\ &= \frac{\sigma^2}{\sum_{i=1}^N x_i^2} = \frac{\text{Var}(\varepsilon)}{\text{Var}(x)}\end{aligned}$$

From this expression, we can build **confidence intervals**.

The above is saying that our beta equals the true beta + error term, which depends on how x and epsilon correlate/covary.

From going to the first line to the second, take the variance of both sides.

x_i are treated like fixed, non-random constants, like a known number. So when you take variance, treat it like a scalar multiplier.

The idea is that you are conditioning on the observed values of x, asking given these x values how much does β vary?

This is the same as assumption 1: $\mathbb{E}(\varepsilon|x) = 0$ — you're always thinking conditionally on x.

ε is treated as random, and the source of all randomness in the model. It's random because it captures unobserved factors that vary unpredictably, different samples would give different ε realisations and x is observed and fixed whereas ε is never observed and always uncertain.

$\hat{\beta}$ is random because it depends on ε — this is the entire reason β has a variance at all.

Continuing:

Since β itself, the true value, is constant, $\text{Var}(\beta)=0$, so all variance in $\hat{\beta}$ comes from the error term.

Now the key result at the bottom is that the more variance in epsilon, the more variance and less precise β .

And just like we touched on in Stats, the more x varies the more precise β because it is doing more work identifying the relationship, x needs to vary to fundamentally find out how much x varies with Y to find the relationship.

We can (now) build confidence intervals because β is normally distributed and we need its standard deviation for this.

Confidence intervals:

A misconception is thinking 95% CI means there's a 95% chance the true β is in this interval. But the true β is a fixed number and either is or is not in the interval. It means that if you repeated the study 100 times with 100 different samples, and built a CI each time, 95 of those intervals would contain the true β .

Confidence level is how reliable the procedure is. This is a probability you choose before, is an input. Is the expected percentage of times a sample falls within a specific range if an experiment is repeated several times.

Regression tables:

	(1) ①
	Log(Monthly Business Revenue) ②
Log(Total Business Assets) ③	0.217 ④ (0.0199)*** ⑤ ← ⑥
Observations	650 ⑦

Observations	650
R-squared	0.16
Standard errors in parentheses	
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$	

1. Regression/column number
2. Dependent variable
3. Explanatory variable(s)
4. Regression coefficient(s)
5. Regression standard error(s)
6. Stars to indicate p -values
7. Other statistics
8. Table notes

What p -values are:

A p -value is the probability of seeing data at least as extreme as what you observed, assuming the null hypothesis is true.

It is NOT the probability that the null hypothesis is true.

$p = 0.03$ means: if the null were true, I'd see a result this extreme only 3% of the time by chance

$p < 0.05 \rightarrow$ statistically significant at 5% level

If you run 20 hypothesis tests, you'd expect 1 to be significant at $p < 0.05$ purely by chance even if nothing is real.

What affects the p -value:

Larger sample $\rightarrow$ smaller standard errors $\rightarrow$ larger t -statistic $\rightarrow$ smaller p -value (see the t -stat as a function of standard error below).

Larger effect size $\rightarrow$ larger t -statistic $\rightarrow$ smaller p -value

More noise in data $\rightarrow$ larger standard errors $\rightarrow$ larger p -value

Standard error, defining this and then going into t -stat:

Measures how much the average of a sample is expected to vary from the true population average. It is the standard deviation of the estimator. Standard error is the standard deviation of how accurate an estimator is across many hypothetical samples.

The issue is that the true standard deviation, σ_β , is never observed and we must estimate it as $SE(\hat{\beta})$.

Note that σ_β is not the standard deviation of the data, it is the true standard deviation of $\hat{\beta}$ itself and how much β varies across repeated samples. So $SE(\hat{\beta})$ and σ_β are both standard deviations of $\hat{\beta}$.

We care about the uncertainty of $\hat{\beta}$ not the raw data, and this is what the hypothesis test is about, so don't use the (sample) standard deviation which is the spread of the raw data.

$$SE(\hat{\beta}) = \hat{\sigma} / \sqrt{\sum x_i^2}$$

This literally just comes from the final part of:

$$\text{Var}(\hat{\beta}) = \frac{\sigma^2}{\sum_{i=1}^N x_i^2} = \frac{\text{Var}(\varepsilon)}{\text{Var}(x)}$$

$\text{Var}(\varepsilon) = \sigma^2$ measures how much the data points scatter around the true regression line.

Large $\text{Var}(\varepsilon) \rightarrow$ data is noisy, lots of unexplained variation $\rightarrow$ harder to pin down β precisely.

Note on the above that when we write $y_i = \beta \cdot x_i + \varepsilon_i$ we assume that the true relationship between x and y is linear and that ε mops up everything else (measurement error, omitted variables (other regressors we are not including which we should), random noise, any non-linear component of the true relationship).

If the non-linearity is systematic, then ε is no longer random noise — it has a pattern correlated with x . This violates assumption 1: $E(\varepsilon | x) = 0$, because the non-linear component in ε depends on x .

To deal with this we would add polynomial terms like X^2 , X^3 to capture curvature, use log transformations and non-parametric methods that don't impose linearity.

Back on topic:

$\text{Var}(X)$ in the above is the spread of the regressor x in the observations. Large $\text{Var}(x) \rightarrow x$ takes a wide range of values $\rightarrow$ gives OLS more "leverage" to identify the slope $\rightarrow$ more precise β .

But replacing the population standard deviation with the sample sd:

Where: $\hat{\sigma}^2 = \sum \hat{\epsilon}_i^2 / (N - K)$

$N-K$ degrees of freedom, N observations and K estimated parameters

So by estimating standard deviation (error) that makes $K+1$

The t-statistic:

T-stats follow the Student's t-distribution. It's a zero-mean symmetrical bell curve with fatter tails than a normal distribution (so it is not normally distributed but looks similar).

If a dataset or variable is t-distributed, it is not normally distributed, though it approaches a normal distribution as the sample size grows. This is because the t-distribution is based on degrees of freedom (typically based on sample size) and when this increases it becomes increasingly identical to the normal dist. Generally when $n > 30$, the difference is negligible between t- and normal-dist.

First, $\hat{\beta}$ is approximately normally distributed, trivially $\hat{\beta} \sim \mathcal{N}(\beta, \sigma_{\hat{\beta}}^2)$. If we standardise (subtract mean, divide by standard deviation) we get: $(\hat{\beta} - \beta) / \sigma_{\hat{\beta}} \sim \mathcal{N}(0,1)$

Under the null hypothesis of $\beta=0$ (that a factor does not have significant effect) this becomes $\hat{\beta} / \sigma_{\hat{\beta}} \sim \mathcal{N}(0,1)$

But, standard error is also not known and has to be estimated (above part).

Now it's no longer exactly standard normal and it follows a t-distribution with $N-K$ degrees of freedom (N observations, K regressors).

The t-distribution has fatter tails reflecting the increased uncertainty in comparison with the normal distribution.

As $N \rightarrow \infty$:

The t-distribution converges to the standard normal.

$SE(\hat{\beta})$ becomes a very accurate estimate of $\sigma_{\hat{\beta}}$.

With that all being said:

$t = \hat{\beta} / SE(\hat{\beta})$

This just measures how many standard errors your estimate is away from zero. The p-value is literally just the area in the tails of that distribution beyond your observed t-statistic.

Small t-stat $\rightarrow$ you're close to zero $\rightarrow$ large tail area $\rightarrow$ large p-value $\rightarrow$ not significant

Large t-stat $\rightarrow$ you're far from zero $\rightarrow$ small tail area $\rightarrow$ small p-value $\rightarrow$ significant

When the p-value is significant:

If your p-value is below your threshold (e.g. 0.05), it means your data is unlikely to have occurred by chance if the null hypothesis were true.

The chain goes:

You assume $H_0: \beta = 0$ (no effect)

You compute $t = \hat{\beta} / SE(\hat{\beta})$

You ask: how likely is a t-stat this large if β really were 0?

If that probability (the p-value) is very small $\rightarrow$ your data is hard to explain under the null $\rightarrow$ you reject H_0

If the null is true, results like the sample would be rare and there is evidence against the null.

Statistical significance tells you about signal relative to noise — it's heavily influenced by sample size. A tiny effect can be significant with huge N . A large effect can be insignificant with tiny N .

Effect size is just how big the effect actually is — the magnitude of β itself, in real world terms.

A drug reduces blood pressure by 0.001 mmHg — highly significant with $N = 1,000,000$, but completely meaningless clinically.

From the table above, the effect size is 0.208 — a 1% increase in assets leads to a 0.208% increase in revenue. That's the economically meaningful quantity. The beta coefficient itself is the effect size, they are the same thing.

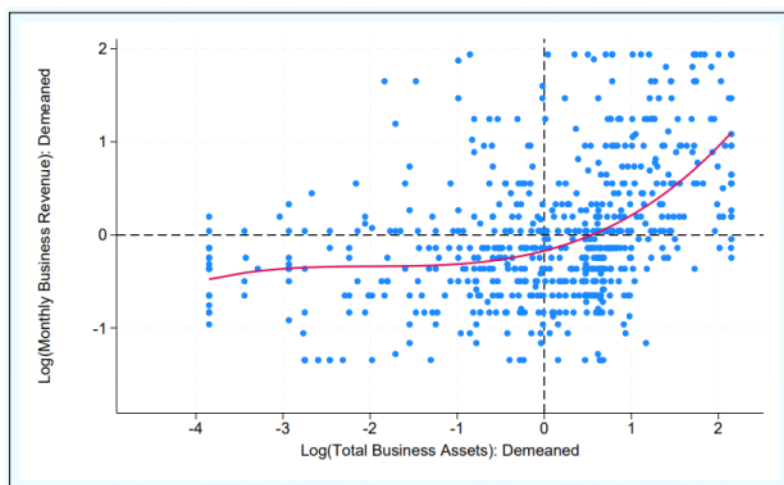
Lecture 2

30 March 2026 20:00

Multiple regressors.

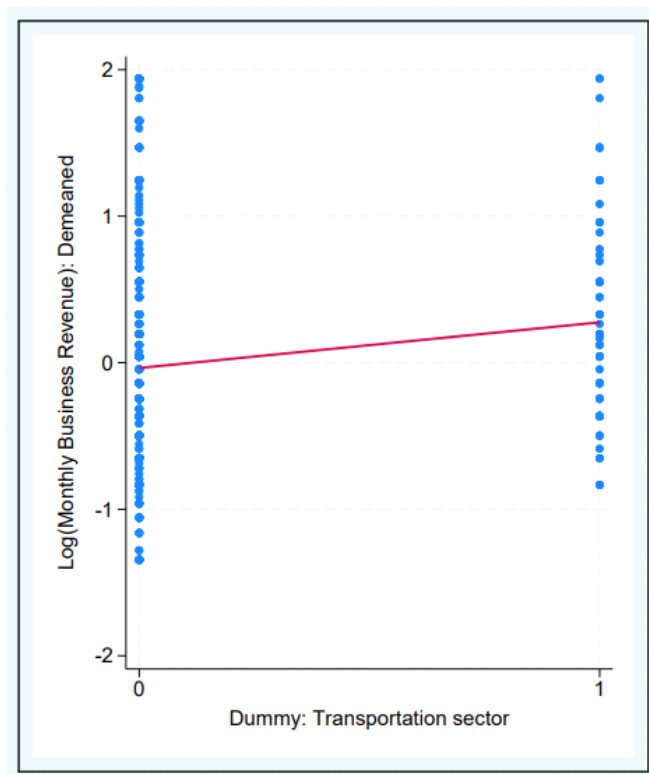
$$y_i = \beta_0 + \beta_1 \cdot x_i + \beta_2 \cdot x_i^2 + \beta_3 \cdot x_i^3 + \varepsilon_i$$
$$(\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3) \equiv \arg \min_{(\beta_0, \beta_1, \beta_2, \beta_3)} \sum_{i=1}^N (y_i - \beta_0 - \beta_1 \cdot x_i - \beta_2 \cdot x_i^2 - \beta_3 \cdot x_i^3)^2$$

Which can make predictions more accurate than just the linear variation



Alternatively, to make a line of best fit you can use a moving average of y_i with respect to x_i , sometimes called kernel regression, but not covered in this module.

A dummy variable takes only 0 or 1 as a value and is used eg to categorise. Eg $D_i = 1$ if respondent i satisfies a certain condition, like being in the transportation sector.



We usually don't actually show this plot though, but this is just visualising the regression of $y_i = \beta_0 + \beta_1 \cdot D_i + \varepsilon_i$

An interaction term is a regressor that is formed as the product of two other regressors.

This allows us, with dummy variables, to estimate alternative model simulations at the same time, like an if statement.

For example, we might say...

$$y_i \equiv \begin{cases} \alpha_0 + \alpha_1 \cdot X_i + \varepsilon_i & \text{if respondent } i \text{ is not in the transportation sector} \\ \alpha_0 + \beta_0 + (\alpha_1 + \beta_1) \cdot X_i + \varepsilon_i, & \text{if respondent } i \text{ is in the transportation sector.} \end{cases}$$

$$\therefore y_i = \alpha_0 + \beta_0 \cdot D_i + \alpha_1 \cdot X_i + \beta_1 \cdot D_i \cdot X_i + \varepsilon_i$$

Activate

Note on notation, normally we have:

$$y = \beta_0 + \beta_1 \cdot X + \varepsilon$$

The beta letters have just been replaced by alpha to let beta letters be used for transport-specific effects.

$$\rightarrow y = \alpha_0 + \alpha_1 \cdot X + \varepsilon$$

In the slide above, we have

$$y = \alpha_0 + \beta_0 \cdot D_i + \alpha_1 \cdot X_i + \beta_1 \cdot D_i \cdot X_i + \varepsilon$$

You can see that β_0 and β_1 only exist if $D_i = 1$

Which is what the above shows.

And hence if they only exist when $D_i = 1$, then their values will be transport specific.

The full interaction model above allowed transport and non-transport workers to have both a different intercept and a different slope, but if you believe only the intercept differs, and the slope is the same, then:

$$\text{You can apply a restriction, from } y = \alpha_0 + \beta_0 \cdot D_i + \alpha_1 \cdot X_i + \beta_1 \cdot D_i \cdot X_i + \varepsilon$$

Drop the interaction term $\beta_1 \cdot D_i \cdot X_i$, this is an interaction term because X_i and D_i are both regressors. It is only switched

on when $D_i = 1$.

D_i is a regressor because a regressor is any variable that explains y .

Now if you plug in $D_i = 1$ into the dropped variation, the slope a_1 is identical for $D_i = 0$ or 1

- Plug in $D_i = 0$: $y = \alpha_0 + \alpha_1 \cdot X_i + \varepsilon$
- Plug in $D_i = 1$: $y = (\alpha_0 + \beta_0) + \alpha_1 \cdot X_i + \varepsilon$

The intercept of constants differs, but the slope a_1 is the same.

Before there was another X_i term, changing the slope, that has now been dropped.

Warning: The 'dummy variable trap'



Suppose that D_i is a dummy for being in the **transportation sector**. Suppose N_i is a dummy for **not** being in the transportation sector. What's wrong with the following model? (How could you fix it?)

$$y_i = \beta_0 + \beta_1 \cdot D_i + \beta_2 \cdot N_i + \varepsilon_i.$$

In this example $B + N = 1$ always, so this is perfect multicollinearity, the regression cannot identify β_0 or β_1 as there are infinite combinations, so no unique solutions. Just drop one of the dummies.

$$y_i = \sum_{k=1}^K \beta_k \cdot X_{ki} + \varepsilon_i$$

Every regression we have seen so far can be written as the above.

You may ask where is the intercept term β_0 but it is secretly already in the formula since:

If you set $X_{1i} = 1$ for every observation (a column of ones), then:

$$\beta_1 \cdot X_{1i} = \beta_1 \cdot 1 = \beta_1$$

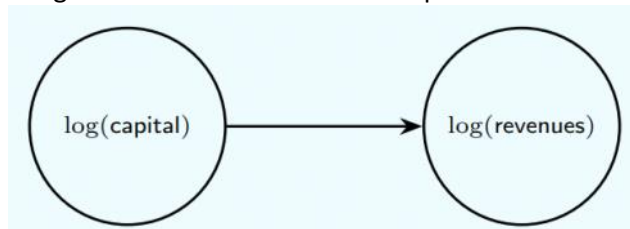
So the general form handles it by:

- $k=1$: $X_{1i} = 1$ (the constant/intercept)
- $k=2$: X_{2i} = your first real regressor
- $k=3$: X_{3i} = your second regressor
- ...and so on up to K regressors

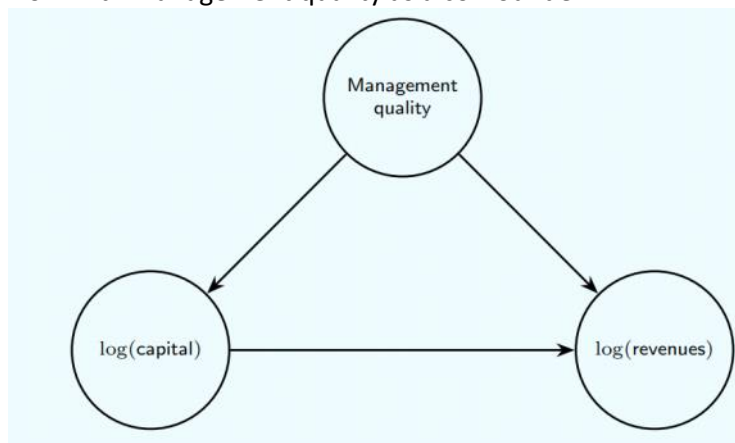
Directed graphs are graphs where every edge has a direction.

A Directed Acyclic Graph (DAG) is a directed graph where no directed paths return to their starting nodes, no cycles.

Imagine a scenario where more capital causes more revenues:



Now with management quality as a confounder:



A Confounder is a variable that affects both the explanatory variable/regressor X and the outcome Y. It makes it look like X causes Y, when actually both are driven by a third variable.

In the above, management quality affects both, so just regressing the bottom two will overestimate the effect of capital on revenues.

This is an issue for OLS because of assumption $E(\epsilon_i | x_i) = 0$, meaning the error term must be uncorrelated with x, but confounders sit in the error term and drives capital, assumption violated, this is endogeneity and you should try pulling out management quality as a variable on its own to control for it. Is the motivation for multiple regressors.

The above describes omitted variable bias.

Endogeneity is when your regressor X is correlated with the error term ϵ .

Suppose we have the following data-generating process (where, for simplicity, x_i , z_i and y_i have all been de-meaned already)...

$$x_i = \alpha_1 \cdot z_i + \mu_i;$$

$$y_i = \beta_1 \cdot x_i + \beta_2 \cdot z_i + \epsilon_i$$

Now suppose we regress y_i on x_i (omitting z_i from the regression). Then we have...

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N x_i \cdot y_i}{\sum_{i=1}^N x_i^2} = \beta_1 \cdot \frac{\sum_{i=1}^N x_i^2}{\sum_{i=1}^N x_i^2} + \beta_2 \cdot \frac{\sum_{i=1}^N x_i \cdot z_i}{\sum_{i=1}^N x_i^2} + \frac{\sum_{i=1}^N x_i \cdot \epsilon_i}{\sum_{i=1}^N x_i^2}$$

$$\therefore \text{plim } \hat{\beta}_1 = \beta_1 + \beta_2 \cdot \frac{\text{Cov}(x, z)}{\text{Var}(x)}$$

Run OLS on y as the outcome and x as the only regressor. We know the true data generating process is:

$$y_i = \beta_1 \cdot x_i + \beta_2 \cdot z_i + \epsilon_i$$

So plug y into the standard OLS formula: $\hat{\beta}_1 = \Sigma x_i y_i / \Sigma x_i^2$

$\Sigma x_i \cdot y_i$ turns into $\Sigma x_i \cdot (\beta_1 \cdot x_i + \beta_2 \cdot z_i + \epsilon_i)$, rearrange and you get the above.

In the first two lines, z is a part of both x and y, it is a confounder.

SIDE NOTE:

Remember that the first part

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N x_i \cdot y_i}{\sum_{i=1}^N x_i^2}$$

Comes from

OLS minimises the sum of squared residuals:

$$\min \sum \hat{\varepsilon}_i^2 = \sum (y_i - \hat{\beta} \cdot x_i)^2$$

Take the derivative with respect to $\hat{\beta}$ and set to zero:

$$d/d\hat{\beta} \cdot \sum (y_i - \hat{\beta} \cdot x_i)^2 = 0$$

$$= \sum 2(y_i - \hat{\beta} \cdot x_i) \cdot (-x_i) = 0$$

Divide through by -2:

$$\sum x_i(y_i - \hat{\beta} \cdot x_i) = 0$$

Expand:

$$\sum x_i y_i - \hat{\beta} \cdot \sum x_i^2 = 0$$

Rearrange for $\hat{\beta}$:

$$\hat{\beta} = \sum x_i y_i / \sum x_i^2$$

Setting the derivative to zero is forcing $\text{Cov}(\hat{\varepsilon}, x) = 0$

BACK TO THE MAIN:

$\hat{\beta}_1$ splits into three parts, first the true effect we want which equals β_1 , the second part the bias from the omitted confounder which we don't want, in red, and third fraction the noise which disappears as $n \rightarrow \infty$

The other terms cancel out, but the covariance between x and z remains, when it shouldn't.

Our $\hat{\beta}_1$ estimator converges to the truth plus a bias term. The red looks like the beta formula because it is, it's the coefficient you'd get from regressing z on x , it literally is just measuring how much the confounder and the regressor move together.

So the bias is β_2 (how much z affects y) . how strong x and z are correlated. If either are zero, no bias.

Back to the idea of effective value and strength of relationship.

Whether to include (in our example) management quality in our regression:

So – should we include management quality as a control?

OUR BELIEF ABOUT MANAGEMENT QUALITY	INCLUDE AS A CONTROL?
management quality as confounder	yes
revenue as common effect	optional
management quality as mediator	no
management quality as collider	no
management quality as downstream effect	no

Only if it's a confounder. Any other, which is just different directions/relationships on DAGs, don't include.

In the 'no' categories, including management quality as a control variable will bias our estimate of the causal effect.

What controlling for management quality (confounder) may look like:

	(1)	(2)
	Log(Monthly Business Revenue)	Log(Monthly Business Revenue)
Log(Total Business Assets)	0.217 (0.0199)***	0.208 (0.0201)***
Management score		0.0783 (0.0276)**
Constant	4.831 (0.138)***	4.893 (0.139)***
Observations	650	650
R-squared	0.16	0.17

Standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

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The coefficients dropped from 0.217 -> 0.208, not much but still slightly inflated before.

Management score is significant **, so it genuinely matters for revenue.

The R^2 barely moves, management adds some but not huge explanatory power.

The standard errors are almost identical, adding management didn't change much precision.

Note that in:

$$\therefore \text{plim } \hat{\beta}_1 = \beta_1 + \beta_2 \cdot \frac{\text{Cov}(x, z)}{\text{Var}(x)}$$

The bias only becomes large if both parts of the second term are large.

β_2 is how much management directly affects revenue (0.0783, fairly small).

The red fraction is how correlated capital and management are.

So even if management and capital are reasonably correlated, if β_2 is small, the whole bias term stays small.

It's a multiplication, so if either part is close to 0, the bias stays small. That's why the coefficient only shifted 0.217 -> 0.208 because management wasn't a powerful enough confounder to reduce more bias.

Lecture 3

31 March 2026 12:47

Stacking regressors:

$$\begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix}_{N \times 1} = \begin{pmatrix} x_{11}, & \dots, & x_{K1} \\ \vdots & \ddots & \vdots \\ x_{1N}, & \dots, & x_{KN} \end{pmatrix}_{N \times K} \cdot \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_K \end{pmatrix}_{K \times 1} + \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_N \end{pmatrix}_{N \times 1}$$
$$\mathbf{y} = \mathbf{X} \cdot \boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

$$\text{In } \mathbf{y} = \mathbf{X} \cdot \boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

The matrix $\mathbf{X}$ is known as the design matrix
Matrix $\mathbf{X}'\mathbf{X}$ is the Gram matrix.

For the design matrix, rows are observations, columns are regressors. Named design matrix as it designs/lays out the structure of the regression.

For the Gram matrix, $\mathbf{X}'$ means $\mathbf{X}$ transposed, this appears from OLS: $\hat{\boldsymbol{\beta}} = \Sigma x_i y_i / \Sigma x_i^2$, where the denominator is $\mathbf{x}$ multiplied by itself, which is what $\mathbf{X}'\mathbf{X}$ is.

The gram matrix is just the matrix version of $\mathbf{X}^2$. And when the different regressors are the columns (like above), then the gram matrix has to be written as $\mathbf{X}'\mathbf{X}$. If it was transposed (the other way around), then it can be written as $\mathbf{X}\mathbf{X}'$.

$$\begin{aligned} \mathbf{y} &= \mathbf{X} \cdot \hat{\boldsymbol{\beta}} + \hat{\boldsymbol{\varepsilon}} \\ \mathbf{X}' \cdot \mathbf{y} &= \mathbf{X}' \cdot \mathbf{X} \cdot \hat{\boldsymbol{\beta}} + \mathbf{X}' \cdot \hat{\boldsymbol{\varepsilon}} \\ \mathbf{X}' \cdot \hat{\boldsymbol{\varepsilon}} &= \mathbf{0} \\ \therefore \mathbf{X}' \cdot \mathbf{y} &= \mathbf{X}' \cdot \mathbf{X} \cdot \hat{\boldsymbol{\beta}} \\ \therefore \hat{\boldsymbol{\beta}} &= (\mathbf{X}' \cdot \mathbf{X})^{-1} \cdot \mathbf{X}' \cdot \mathbf{y}. \end{aligned}$$

In the above, $\mathbf{X}$ is not a square matrix

This is just the matrix version of what we have seen before with scalars.

For the above, first, $\mathbf{y}$ and $\mathbf{x}$ have no hats because they are actual observed data.

Beta and epsilon have hats because these are estimated, we never observe the true values, only estimate them from the data.

Third line is the orthogonality condition, where residuals are uncorrelated with every regressor. This is imposed by OLS, in the derivation of $\hat{\boldsymbol{\beta}}$ when you differentiate and $\min \boldsymbol{\beta}$, the part $-2\mathbf{X}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = 0$ is just $\mathbf{X}'\hat{\boldsymbol{\varepsilon}} = 0$. Then rearrange.

In scalar algebra, if you have $ax = b$, then $x = a^{-1}b$, simple.

In matrix algebra, for OLS: $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$

We try to do the same thing by 'dividing' both sides by X to solve β hat. The issue is that X is not a square matrix (more observations than regressors) so you can't directly invert it. Only square matrices are directly invertible.

Let's introduce X tilda, $\tilde{X} = (X'X)^{-1}X'$

This behaves like an inverse even when the true inverse does not exist. Called a pseudo-inverse.

1. $\tilde{X} \cdot X = (X' \cdot X)^{-1} \cdot X' \cdot X = I$.
2. If X is an invertible square matrix, $(X' \cdot X)^{-1} \cdot X' = X^{-1} \cdot (X^{-1})' \cdot X' = X^{-1}$.

Note that when a matrix is multiplied by its transposed self, it is square and symmetric. Here, this is the Gram matrix, capturing all inner products.

Remember that you cannot do matrix multiplication in any way you want, what's allowed:

Associativity: $(A \cdot B) \cdot C = A \cdot (B \cdot C)$

Distributivity: $A \cdot (B + C) = A \cdot B + A \cdot C$

But NOT: $A \cdot B = B \cdot A$

For the shapes of two matrices being multiplied, the inner dimensions, eg $m \times n$ and $n \times p$, must match. So it's not quite matrix multiplication is left to right, it is using associativity and distributivity, above.

The inverse of a transpose is the same thing as the transpose of the inverse: $(X^{-1})' = (X')^{-1}$

So: $(X^{-1})' \cdot X' = (X')^{-1} \cdot X' = I$

Hence the last two terms in the second result cancel when we substitute one of them into the other. We can cancel the last two (in terms of multiplication 'order' because they are next to each other and the result is a standard result which definitely has compatible matrix dimensions for multiplication.

You can multiply the last two together because associativity allows you to bracket any adjacent pair.

Remembering that $(AB)^{-1} = B^{-1}A^{-1}$

Point 1 shows that $\tilde{X} \cdot X$ is the identity matrix, like an inverse.

Point 2 says when X itself is square, so has an inverse, the RHS $(X'X)^{-1}X'$ collapses to X^{-1} exactly.

$\hat{\beta} = (X' \cdot X)^{-1} \cdot X' \cdot y$ has some very nice properties. In particular, these include...

1. $\hat{\beta}$ is **unbiased** (in fact, it is the '**Best Linear Unbiased Estimator**' for β);
2. $\hat{\beta}$ is **consistent**: $\text{plim } \hat{\beta} = \beta$;
3. The Central Limit Theorem applies (so $\hat{\beta}$ has an asymptotically Normal distribution); and
4. $\hat{\beta}$ has a known **variance** (this relates closely to the **inverse of the Gram matrix**, but we will not discuss this in any detail).

For a matrix A to be invertible, if $A \cdot v = 0$, then $v = 0$

When does $\hat{\beta} = (X' \cdot X)^{-1} \cdot X' \cdot y$ exist?

$\hat{\beta}$ exists if $(X' \cdot X)$ is non-singular. If X has 'reduced rank' – that is, the columns are linearly dependent – we can find some vector $v_{K \times 1} \neq 0$ such that:

$$X \cdot v = 0.$$

Now, let's multiply both sides of this equation by X' :

$$(X'X) \cdot v = 0.$$

This is the definition of $X'X$ being **singular**.

Key concept: If the design matrix has reduced rank, $\hat{\beta}$ does not exist.

If the design matrix (X) has reduced rank, then the Gram matrix ($X'X$) is singular, and $\hat{\beta}$ does not exist.

What the above means:

Singular = non-invertible

Invertible = non-singular

A matrix A is invertible if A^{-1} exists such:

$$A \cdot A^{-1} = A^{-1} \cdot A =$$

You cannot invert a singular matrix, you the Gram matrix needs to be invertible. We examine X to check if it is invertible.

A reduced rank is when the columns are linearly dependent, meaning one column can be written as a combination of others. Eg when there are two dummy variables and one equals 1 when the other is 0 and vice versa.

If the columns are linearly dependent, there exists a non-zero vector v such $X \cdot v = 0$

And hence multiply both sides by X' , then $X'X \cdot v = 0$, definition of non-singularity (invertibility), so $X'X$ is singular and has no inverse.

A matrix is not invertible when it contains redundant information. It basically says that the matrix does not have enough independent information to uniquely determine a solution.

It's like if you had the equation $0x = 0$, this doesn't allow us to solve for x .

Hence $\hat{\beta}$ does not exist if $(X'X)^{-1}$ does not exist.

This all just shows you why you can't have that dummy variable trap where they are functions of each other.

The opposite is true, where if $\hat{\beta}$ does not exist, it must be because the Gram matrix is singular, implying the design matrix X has reduced rank.

The Frisch-Waugh-Lovell ('FWL') theorem:

Let's think, in matrix terms, about the **predicted values**:

$$\begin{aligned}\hat{y} &= X\hat{\beta} \\ &= X \cdot (X' \cdot X)^{-1} \cdot X' \cdot y \equiv P_X \cdot y.\end{aligned}$$

And now let's think about the **residuals**:

$$\begin{aligned}y - \hat{y} &= y - X\hat{\beta} \\ &= y - P_X \cdot y \\ &= [I - P_X] \cdot y \equiv M_X \cdot y.\end{aligned}$$

Think of $\hat{y}$ as the part of y our regressors X can explain. P_X is the matrix that projects y onto the space spanned by X , so extracts the explainable part.

The penultimate line says that the residual is what's left from $y - P_X \cdot y$ (the parts we cannot explain). So, explicitly, $M_X \cdot y$ gives you the residuals

Think of y being split into perpendicular planes, $P_X \cdot y$ are the fitted values, $M_X \cdot y$ is the remainder, residuals, at a right angle. Projection matrix P_X and the annihilator matrix M_X .

$$M_X \equiv I - X \cdot (X' \cdot X)^{-1} \cdot X'$$

1. $M_X \cdot X = 0$.
2. M_X is a **symmetric** $N \times N$ matrix.
3. M_X is **idempotent**: $M_X \cdot M_X = M_X$.

M_X is the annihilator matrix because it multiplied by $X = 0$, destroys it (see why below). It destroys any variation in y that can be explained by X , leaving only the residuals behind.

2 is a technical property

3 shows how once you have removed the X component from y , removing it again does nothing.

We have $y = X \cdot \beta_1 + Z \cdot \beta_2 + \varepsilon$

Where z are the controls we want to account for.

$$M_Z \cdot y = M_Z \cdot X \cdot \beta_1 + M_Z \cdot \varepsilon.$$

Define by M_Z the annihilator matrix with respect to Z . Let's multiply all terms by M_Z .

Now, we know that $M_Z \cdot Z = 0$, so we can simplify.

So, instead of regressing y on X and Z , we could:

1. Regress y on Z and save the residuals;
2. Regress X on Z and save the residuals;
3. Regress the residualised y (that is, $M_Z \cdot y$) on the residualised X (that is, $M_Z \cdot X$).

The logic on this slide shows that these are the same **model** – but this does not (yet) show us that the two approaches will give **identical estimates**...

Multiply each term by M_Z , Z disappears.

$M_Z \cdot y$ are the residuals of y after removing z 's influence.

$M_Z \cdot X$ are the residuals of x after removing z 's influence.

So you are left with a regression of both variables cleaned of z . Regress these together gives us β_1 . This basically just shows us why controlling for Z actually works.

What the third step looks like:

3. Regress the residualised y (that is, $M_Z \cdot y$) on the residualised X (that is, $M_Z \cdot X$).

What does the estimator $\hat{\beta}_1$ look like?

$$\begin{aligned}\hat{\beta}_1 &= [(M_Z \cdot X)' (M_Z \cdot X)]^{-1} \cdot (M_Z \cdot X)' \cdot (M_Z \cdot y) \\ &= (X' \cdot M_Z' \cdot M_Z \cdot X)^{-1} \cdot X' \cdot M_Z' \cdot M_Z \cdot y \\ &= (X' \cdot M_Z \cdot X)^{-1} \cdot X' \cdot M_Z \cdot y.\end{aligned}$$

Replacing X with $M_Z \cdot X$ and y with $M_Z \cdot y$ from the standard matrix beta estimator formula.

What we will find out is that we can actually just regress the original y on the residualised X ($M_Z \cdot X$).

This gives the same $\hat{\beta}_1$.

M_Z only needs to be applied once to y because $X' \cdot M_Z = (M_Z \cdot X)'$, so the M_Z on y gets absorbed naturally.

Define $\hat{\alpha}$ as the OLS coefficients from regressing each column of X on the matrix Z : $P_Z \cdot X \equiv Z \cdot \hat{\alpha}$. Let's rewrite our original model (remembering that $P_Z + M_Z = I$):

$$\begin{aligned}y &= X \cdot \beta_1 + Z \cdot \beta_2 + \epsilon \\ &= (M_Z + P_Z) \cdot X \cdot \beta_1 + Z \cdot \beta_2 + \epsilon \\ &= M_Z \cdot X \cdot \beta_1 + P_Z \cdot X \cdot \beta_1 + Z \cdot \beta_2 + \epsilon \\ &= M_Z \cdot X \cdot \beta_1 + Z \cdot (\hat{\alpha} \cdot \beta_1 + \beta_2) + \epsilon\end{aligned}$$

Now, $M_Z \cdot X$ is orthogonal to Z – so we will obtain identical $\hat{\beta}_1$ whether we include Z in this regression or not. It follows that we can omit Z and simply write:

$$\hat{\beta}_1 = (X' \cdot M_Z' \cdot M_Z \cdot X)^{-1} \cdot X' \cdot M_Z' \cdot y = (X' \cdot M_Z \cdot X)^{-1} \cdot X' \cdot M_Z \cdot y$$

This slide just extends beyond the above idea.

$M_Z + P_Z = 1$ since this is the entire ratio of every part of x is explained by z or not.

You're left with $y = M_Z \cdot X \cdot \beta_1 + Z \cdot (\hat{\alpha} \cdot \beta_1 + \beta_2) + \epsilon$

Now $M_Z \cdot X$ is orthogonal to Z by the annihilator property. So Z carries no information about β_1 — including or excluding Z makes no difference to $\hat{\beta}_1$.

This proves FWL formally: you get identical $\hat{\beta}_1$ whether you run the full regression or just regress residualised y on residualised X .

From the line above the text, you conclude that you don't have to include z , just regressed on X with original Y . So take the standard OLS formula, $(X'X)^{-1}X'y$ but replacing X with $M_Z \cdot X$ and use the symmetry and idempotency properties to simplify.

$(M_Z \cdot X)' = X' \cdot M_Z' = X' \cdot M_Z$ (symmetry)

$M_Z \cdot M_Z = M_Z$ (idempotency)

Giving: $\hat{\beta}_1 = (X' \cdot M_Z \cdot X)^{-1} \cdot X' \cdot M_Z \cdot y$

Demeaning as an application of FWL...

Suppose we have:

$$y = X \cdot \beta_1 + Z \cdot \beta_2 + \epsilon,$$

where Z is a vector of ones (and X does **not** include the usual column of ones). Then..

$$M_Z \equiv I - Z \cdot (Z' \cdot Z)^{-1} \cdot Z'$$

$$= \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix} - \frac{1}{N} \cdot \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 1 & 1 & \cdots & 1 \\ 1 & 1 & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & 1 & 1 \\ 1 & 1 & \cdots & 1 & 1 \end{pmatrix}$$

Multiplying a variable by this M_Z demeans that variable!

$Z \cdot \beta_2$ is just the intercept term, constant. This is because Z here is just a column of ones.

This just says that:

Take any variable x . $M_Z \cdot x$ gives:

$$x - (1/N) \cdot (\text{sum of all } x \text{ values}) = x - \bar{x}$$

That is just demeaning.

By FWL, including an intercept in a regression is equivalent to first demeaning all your variables.

The general takeaway for all regressions:

In other words, regular FWL, applied to the intercept -- controlling for a constant = demeaning everything. This always happens automatically in any regression with an intercept.

Lecture 4

31 March 2026 16:24

Randomised Controlled Trials (RCTs):

First, if you offer microfinance to a group of companies, and try to measure the effect between those who take it and those who don't, that wouldn't be great because those who take it may have better business opportunities, higher motivation, etc. so you cannot tell just because of the loan if microfinance borrowers were already more likely to succeed.

In a randomised controlled trial, the 'treatment' is the policy intervention that is being tested.

The treatment effect / causal effect in an experiment or quasi-experiment.

Treatment group receive the intervention, control group don't.

In a randomised controlled trial, a 'balance table' is a table of covariates, measured at baseline, that shows the similarity (or dissimilarity) of the treatment and control groups. Ideally, the balance table should show no large differences between the groups.

E.g.:

Variable	(1) Control Mean/SE	(2) Treatment: fixed Mean/SE	(3) Treatment: flexible Mean/SE	(4) Total Mean/SE	Normalized difference		
					(1)-(2)	(1)-(3)	(2)-(3)
Age	37.10 (0.65)	37.97 (0.61)	38.65 (0.67)	37.90 (0.37)	-0.09	-0.15	-0.07
Female	0.06 (0.01)	0.10 (0.02)	0.09 (0.02)	0.08 (0.01)	-0.15	-0.13	0.03
Household size	6.11 (0.15)	6.35 (0.21)	6.49 (0.16)	6.31 (0.10)	-0.08	-0.16	-0.05
Household earners	1.91 (0.07)	1.93 (0.08)	2.05 (0.07)	1.96 (0.04)	-0.02	-0.14	-0.10
Distance to MFI office (minutes)	16.14 (1.28)	15.00 (0.91)	14.72 (1.00)	15.29 (0.62)	0.06	0.08	0.02
Business experience (years)	9.31 (0.49)	9.75 (0.47)	9.78 (0.56)	9.61 (0.29)	-0.06	-0.06	-0.00
Number of businesses managed	1.22 (0.03)	1.19 (0.03)	1.25 (0.03)	1.22 (0.02)	0.06	-0.05	-0.11

Estimating the effects of our experiment

We pool surveys from 3 months, 6 months, 12 months, 18 months and 24 months.

Our primary estimating specification is ANCOVA with strata dummies, pooling across treatments and clustering at the respondent level:

$$y_{it} = \beta_0 + \beta_1 \cdot M_i + \beta_2 \cdot y_{i0} + \phi_{s_i} + \varepsilon_{it}.$$

Warning: Imperfect compliance



In this approach, M_i is a dummy variable for having been **offered** the microfinance contract (not a dummy variable for having **accepted** it). (**Why?**)

ANOVA (without C) means Analysis of Variance, just comparing the average outcomes across groups, eg is the average y different between the microfinance group and the control?

A covariate is just a control variable, an additional regressor to account for other factors that affect y.

ANCOVA = ANOVA + covariates, now control for other variables that might affect the outcome.

A plain comparison of treated vs untreated means would be:

$$y = \beta_0 + \beta_1 \cdot M_i + \varepsilon$$

ANCOVA adds covariates on top

$$y_{it} = \beta_0 + \beta_1 \cdot M_i + \beta_2 \cdot y_{i0} + \phi s_i + \varepsilon_{it}$$

- **y_{it}** = outcome for person i at time t (e.g. their revenue)
- **β₀** = intercept/baseline constant
- **β₁·M_i** = the treatment effect — M_i is 1 if offered microfinance, 0 if not. β₁ is what we want to estimate
- **β₂·y_{i0}** = controlling for each person's **baseline outcome** before the experiment started — accounting for where they began
- **φs_i** = strata dummies — the randomisation was done within subgroups (strata), so you control for which subgroup each person belongs to
- **ε_{it}** = error term

The 'fundamental problem of causal inference' is that the researcher can never observe Y_i(1) and Y_i(0) for the same individual i.

In the potential outcomes framework, the Average Treatment Effect is the population average of the individual causal effects is a heterogeneous population — that is, E [Y_i(1) – Y_i(0)]

Empirical CDF refers to using real data rather than a theoretical distribution.

Random definitions I don't really need to know, but I'll just record here:

Inferences and conclusions from an econometric study are internally valid if they are valid for the population being studied.

Inferences and conclusions from an econometric study are externally valid if they can be generalised from the population and the setting being studied to other populations and settings.

The Stable Unit Treatment Value Assumption ('SUTVA') requires that each respondent's outcome is affected only by its own treatment — not by the treatment of others.

The Hawthorne effect is the phenomenon that study participants change their behaviour because they know that their behaviour is being measured.

A John Henry effect occurs when individuals in the control group make extra effort, knowing that they are being compared to individuals in the treatment group.

An Adaptive RCT is an RCT where the probability of treatment changes, based upon the outcomes of previous participants in the study.

Lecture 5

31 March 2026 17:48

Instrumental variables

Ng! this lecture and all the next in the module don't seem that relevant for quant but I am going to go as fast as I can while understanding the content.

An instrumental variable is a variable that causes variation in an endogenous regressor (instrument relevance) and is uncorrelated with the regression error (instrument validity).

In other words, when we say X is endogenous, it means X is correlated with the error term.

Eg X is whether someone attends Hajj, and ϵ contains religiosity (unobserved). Highly religious people both attend Hajj more AND have different attitudes regardless of Hajj

So X and ϵ move together, correlated. So if you want to estimate the effect of X on y , but X is endogenous, use an IV, which has:

Relevance: Z causes variation in X

Validity: Z only affects y through X , not directly. So Z is uncorrelated with ϵ .

People who are more religious both attend Hajj and have different attitudes to religion.

So the instrument of winning the lottery (Z) is relevant (causes people to go to the Hajj) and valid (winning the lottery is random and only affects attitudes through going to the Hajj, not directly).

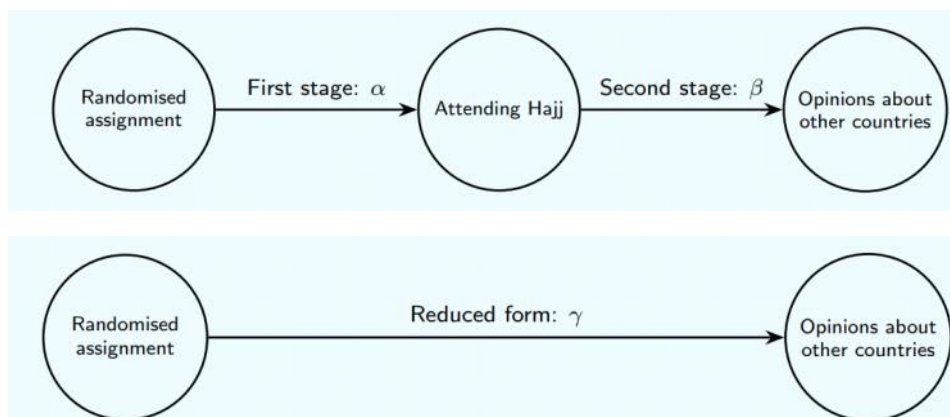
Let's define the following variables:

- y_i attitudes towards other countries;
- x_i a dummy for having attended Hajj;
- z_i a dummy for having won the Hajj lottery.

$$x_i = \alpha \cdot z_i + \mu_i \quad (\text{'First Stage'})$$

$$y_i = \beta \cdot x_i + \varepsilon_i \quad (\text{'Second Stage'})$$

$$y_i = \gamma \cdot z_i + \eta_i \quad (\text{'Reduced Form'})$$



However, if we have a valid instrumental variable, we can impose orthogonality between the second-stage residual (ϵ) and the instrumental variable (z). . .

STOPPED HERE ON PAGE 35 OF LECTURE 5, BECAUSE THE REST of the module looks too pure econometrics and I don't want to waste time and motivation, can always come back to it.

Moving on to Econometrics 2.